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Quantitative Research Correlation
6/21/2007

Instructions
Complete each of the following problems with a simple hand calculator. Use the method demonstrated in class or in the online study unit. Organize your work into columns. Please show all steps of your work. Feel free to use additional sheets of paper.

1. Using the method discussed in class or the online study unit, find the correlation between age and scores on the BSI (Business Skills Inventory). Show all steps of your work. Report the correlation and, in a few plain English sentences, tell what the result means.

Given:

subject #	Age	BSI score
1	50	115
2	40	110
3	40	113
4	30	100

Set of Ages = X = {50, 40, 40, 30}
Set of BSI scores = X = {115, 110, 113, 100}
Number of Ages = N = **4**
Number of BSI scores = N = **4**

1.1 The Mean of Age

Formula: $X' = \sum X / N$

The Mean of Age = X' = $\sum$ (50 + 40 + 40 + 30) / 4 = 160/4 = **40**

1.2 The Mean of BSI scores

Formula: $X' = \sum X / N$

The Mean of BSI scores = X' = $\sum$ (115+110+113+100) / 4 = 438/4 = **109.5**

1.3 Variance

Formula: $V= \sum(X- X')^2 / N$

1.3.1 Variance of Ages

subject #	Age	(X-X')	(X-X')^2
1	50	10	100
2	40	0	0
3	40	0	0
4	30	-10	100

The V of Age = $\sum$ (100 + 0 + 0 + 100) / 4 = 200/4 = **50**

1.3.2 Variance of BSI scores

subject #	BSI score	(X-X')	(X-X')^2
1	115	5.5	30.25
2	110	.5	.25
3	113	3.5	12.25
4	100	-9.5	90.25

The V of BSI score = $\sum$ (30.25 + .25 + 12.25 + 90.25) / 4 = 133/4 = **33.25**

1.4 Standard Deviation

Formula : SD = sqrt(V)

1.4.1 Standard Deviation of Ages

The SD of Ages = sqrt(50) = **7.07**

1.4.2 Standard Deviation of BSI score

The SD of BSI score = sqrt(33.25) = **5.77**

1.5 z score

<u>Formula:</u> z score = z = (X – X’) / SD

1.5.1 The z score of Ages

The mean of ages = 40
The SD of ages = 7.07

subject #	Age	(X-X’)	z score
1	50	10	+1.41
2	40	0	0
3	40	0	0
4	30	-10	-1.41

1.5.2 The z score of BSI scores

The mean of BSI scores = 109.5
The SD of BSI scores = 5.77

subject #	BSI score	(X-X’)	z score
1	115	5.5	+.95
2	110	.5	+.09
3	113	3.5	+.61
4	100	-9.5	-1.65

1.6 The Cross Product of z scores

	X		Y		Cross Product
	1	**2**	**3**	**4**	**5**
subject #	**Age**	**z score**	**BSI score**	**z score**	**z(x) * z(y)**
1	50	+1.41	115	+.95	+1.34
2	40	0	110	+.09	0
3	40	0	113	+.61	0
4	30	-1.41	100	-1.65	+2.33

1.7 Statistic r

<u>*Formula:*</u> $r = \sum(z(x) * z(y)) / N$

r = $\sum(1.34 + 0 + 0 + 2.33)/4 = +3.64/4 =$ **+.918**

1.7.1 What does the correlation tell you (in plain English)?

The correlation of +.918 indicates a strong positive relationship between these variables, such that those who are older in age tend to receive higher BSI *scores.*

1.7.2 What is unusual about the contributions of subjects 2 and 3 to the correlation? Why?

The statistic r of subjects 2 and 3 is zero and it indicates no overall relationship between the people ages and their BSI scores.

2. Using the method discussed in class, or in the online study unit, find the correlation between annual income (expressed in thousands of US dollars) and the number of trips taken on behalf of a school per month. Show all steps of your work. Report the correlation and, in a few plain English sentences, tell what the result means.

<u>*Given:*</u>

Income (in thousands of US dollars)	Number of sponsored trips
50	7
48	6
46	6
30	5
20	4

Set of income = X = {50, 48, 46, 30, 20}
Set of sponsored trips = X = {7, 6, 6, 5, 4}
Number of incomes = N = **5**
Number of sponsored trips = N = **5**

Note: To calculate the statistic r, we need to find the mean, standard deviation, and z-score for each raw score of the two variables to be correlated.

2.1 Mean

<u>*Formula:*</u> X' = ∑ X / N

2.1.1 The mean of income

The mean of income = X' = ∑ (50 + 48 + 46 +30 + 20)/5 = 194/5 = **38.8**

2.1.2 The mean of sponsored trips

The mean of sponsored trips = X' = ∑ (7 + 6 + 6 + 5 + 4)/5 = 28/5 = **5.6**

2.2 Variance

Formula: V= ∑(X- X')^2 / N

2.2.1 The variance in income

Income	(X-X')	(X-X')^2
50	11.2	125.44
48	9.2	84.64
46	7.2	51.84
30	-8.8	77.44
20	-18.8	353.44

The V of income = ∑ (125.44+84.64+51.84+77.44+353.44)/4 = 692.8/5 = **138.56**

2.2.2 The variance of sponsored trips

Sponsored trips	(X-X')	(X-X')^2
7	1.4	1.96
6	.4	.16
6	.4	.16
5	-.6	.36
4	-1.6	2.56

The V of sponsored trips = ∑ (1.96+.16+.16+.36+2.56) / 4 = 5.2/5 = **1.04**

2.3 Standard Deviation

Formula : SD = sqrt(V)

2.3.1 The Standard Deviation of Income

The SD of Income = sqrt(138.56) = **11.77**

2.3.2 The Standard Deviation of Sponsored trips

The SD of sponsored trips = sqrt(1.04) = **1.02**

2.4 z score

Formula: z score = z = (X – X') / SD

2.4.1 The z score of Income

The mean of income = 38.8
The SD of income = 11.77

Income	**(X-X')**	**z score**
50	11.2	+.95
48	9.2	+.78
46	7.2	+.61
30	-8.8	-.75
20	-18.8	-.1.6

2.4.2 The z score of Sponsored trips

The mean of sponsored trips = 5.6
The SD of sponsored trips = 1.02

Sponsored trips	**(X-X')**	**z score**
7	1.4	+1.37
6	.4	+.39
6	.4	+.39
5	-.6	-.59
4	-1.6	-1.57

2.5 Cross Product of z scores

X		Y		Cross Product
1	2	3	4	5
Income	z score	Sponsored trips	z score	z(x) * z(y)
50	+.95	7	+1.37	+1.30
48	+.78	6	+.39	+.30
46	+.61	6	+.39	+.24
30	-.75	5	-.59	+.44
20	-.1.6	4	-1.57	+2.51

2.6 Statistic r

Formula: $r = \sum(z(x) * z(y)) / N$

r = $\sum(1.30 + .30 + .24 + .44 + 2.51)/5 = +4.79/5 =$ **+.958**

What does the correlation tell you (in plain English)?

The correlation of +.958 indicates a strong, positive relationship between these variables, such that higher income brackets tend to take more sponsored trips per month on school behalf.